

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 2 – Seminar on Emerging Technologies

Course Code:	CSA10 / CSA1063	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 2 – Seminar on Emerging Technologies
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
Student Name:	Joanathan Packia Singh	Reg. No.:	192472229
Date:	22/08/2026	Duration:	60 minutes

Code of Conduct

I Joanathan Packia Singh(192472229) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Seminar on Emerging Technologies

Instructions to Students:

Students must select the topic **strictly according to the Serial Number (S.No.) order** assigned to them.

S.No.	Seminar Topic
1	Generative AI and Large Language Models (LLMs)
2	AI Agents and Autonomous AI Systems
3	Multimodal Artificial Intelligence
4	Explainable and Trustworthy AI
5	Edge AI and Intelligent Edge Computing
6	Federated Learning and Privacy-Preserving AI
7	Deep Learning for Computer Vision
8	AI-Powered Medical Image Analysis
9	Natural Language Processing with Transformer Models
10	Retrieval-Augmented Generation (RAG)
11	AI in Cybersecurity and Threat Detection
12	Zero Trust Architecture and Modern Cybersecurity
13	Post-Quantum Cryptography
14	Quantum Computing and Quantum Algorithms
15	Quantum Machine Learning
16	Cloud-Native Computing and Microservices

17	Serverless Computing and Function-as-a-Service
18	Edge Computing and Fog Computing
19	Cloud Security and Confidential Computing
20	Digital Twins and Intelligent Simulation
21	Internet of Things (IoT) and Smart Systems
22	Industrial Internet of Things (IIoT)
23	5G/6G Networks and Next-Generation Communication
24	Blockchain Technology and Decentralized Applications
25	Web3 and Decentralized Internet
26	Smart Contracts and Blockchain Applications
27	Augmented Reality (AR) and Virtual Reality (VR)
28	Extended Reality (XR) and Immersive Technologies
29	Robotics and Autonomous Robots
30	Human–Robot Interaction and Collaborative Robots
31	Autonomous Vehicles and Intelligent Transportation Systems
32	Smart Cities and AI-Driven Urban Technologies
33	Metaverse Technologies and Virtual Worlds
34	Brain–Computer Interfaces (BCI)
35	Wearable Computing and Smart Wearable Devices
36	Green Computing and Sustainable IT
37	AI for Software Engineering and Code Generation
38	DevSecOps and Intelligent CI/CD Pipelines
39	Digital Forensics and AI-Based Cybercrime Investigation
40	Privacy-Enhancing Technologies and Data Protection

Seminar Presentation Structure:

- 1. Introduction to the Emerging Technology**
- 2. Need and Motivation**
- 3. Working Principle / Architecture**
- 4. Key Components and Technologies**
- 5. Applications / Real-World Use Cases**
- 6. Advantages and Limitations**
- 7. Recent Developments and Trends**
- 8. Challenges and Future Scope**
- 9. Case Study / Example**
- 10. Conclusion and References**

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Technical Content & Subject Knowledge (25%)	Comprehensive and accurate content; demonstrates excellent understanding of the emerging technology	Good technical understanding with minor gaps	Basic understanding with limited technical depth	Incomplete content or major technical errors
Organization & Presentation Structure (15%)	Excellent logical flow with clear introduction, content, conclusion, and effective transitions	Well-structured with minor organizational issues	Adequately structured but some sections lack clarity	Poorly structured and difficult to follow
Examples, Diagrams & Case Study (20%)	Excellent use of relevant diagrams, examples, applications, and real-world case studies	Good use of relevant examples and diagrams	Limited examples/diagrams with basic explanation	Inappropriate or insufficient examples/diagrams
Communication & Presentation Skills (20%)	Very clear, confident, engaging delivery with excellent eye contact, voice modulation, and time management	Clear and confident presentation with minor issues	Understandable but lacks confidence or engagement	Unclear delivery, excessive reading, or poor time management
Question Handling & Overall Performance (20%)	Answers questions accurately and confidently; demonstrates strong overall knowledge	Answers most questions correctly	Answers basic questions with some difficulty	Unable to answer questions or demonstrate adequate understanding

Criteria	Mark
Technical Content & Subject Knowledge (25%)	/5
Organization & Presentation Structure (15%)	/5
Examples, Diagrams & Case Study (20%)	/5
Communication & Presentation Skills (20%)	/5
Question Handling & Overall Performance (20%)	/5
TOTAL	/25



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CO5 ASSESSMENT TOOL 2
Seminar on Emerging Technologies
Industrial Internet of Things (IIoT)

CSA1016 – Software Engineering

PRESENTED BY:

JOANATHAN PACKIA SINGH (192472229)

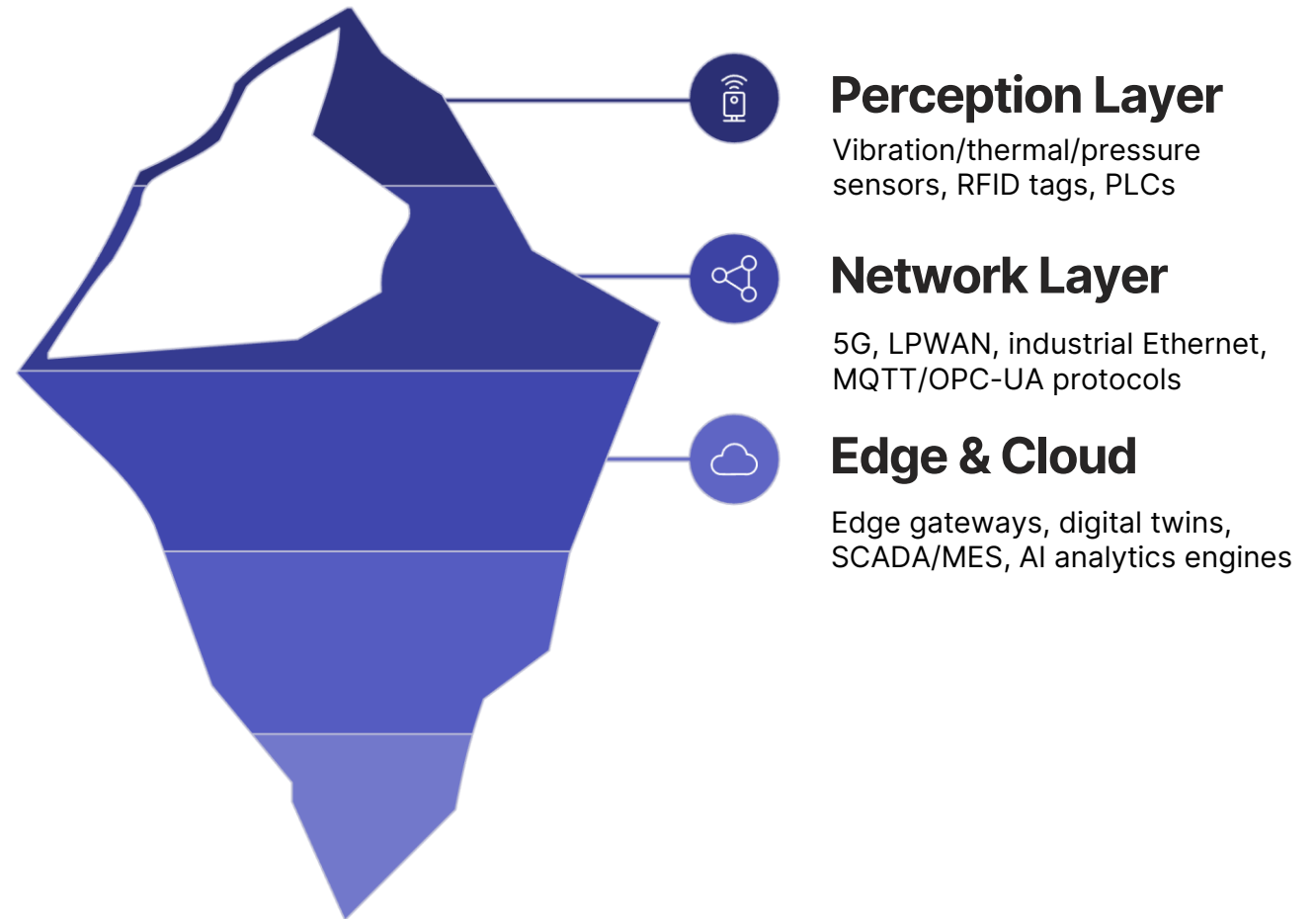
SUPERVISED BY:

DR.K.ANITA DAVAMANI

Industrial Internet of Things (IIoT)

Architecture — From Sensor to Enterprise

- Connecting machines, data, and intelligence to transform operations.
- This briefing explains why IIoT matters for industrial engineers, operations leaders, and technology decision-makers—covering architecture, real use cases, trade-offs, and what's next.



The stack emphasizes low-latency local decisioning at the edge, resilient transport, and enterprise analytics for long-term optimization.

Typical flow: sensors → edge preprocessing → anomaly detection → alert or autonomous actuator command → enterprise records & model retraining.

Why IIoT Matters — The Operational Case

Faster detection

Sensors and edge analytics catch anomalies in minutes instead of weeks, reducing reactive firefighting.

Lower downtime

Up to 45%+ of unplanned downtime is preventable — predictive maintenance cuts failures and lost production.

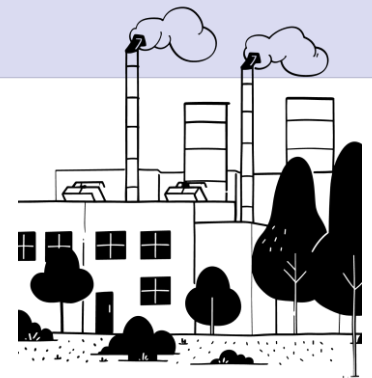
Operational visibility

Real-time KPIs across assets, lines, and sites enable data-driven scheduling and capacity planning.

Safety & remote ops

Remote monitoring and autonomous controls reduce exposure in hazardous environments.

Market signal: global IIoT market $\approx$ \$514B (2025) with aggressive multi-year growth—clear incentive for strategic investment.



Applications, Benefits & Practical Trade-offs

Consumer IoT



Industrial IoT

Key Applications

- Predictive maintenance — reduce MTTR and prevent bearing/motor failures
- Smart factories & Industry 4.0 — adaptive production and quality control
- Supply chain & asset visibility — track location, condition, and utilization
- Energy optimization & smart grids — demand response and load balancing

Advantages

- Reduced downtime and operating cost
- Improved safety and remote operations
- Data-driven process improvements and yield gains

Limitations

- High integration and retrofit costs for legacy assets
- Interoperability gaps across vendors and protocols
- Increased cyber attack surface — need for OT/IT security strategy

Case Study — Siemens Amberg Electronics Plant (EWA), Germany

Closing the Loop: Sensor → Edge AI → Digital Twin → Production Action

- The Challenge: Amberg produces SIMATIC industrial controllers at high volume; an unexplained milling-spindle fault was causing unplanned downtime during PCB depaneling, with no clear root cause.
- The IIoT Solution: Engineers isolated two telemetry signals — spindle rotational speed and drive electric current — and streamed them to an edge device running a pretrained anomaly-detection algorithm
- Technology Stack: Industrial Edge devices + AI anomaly detection → Siemens Insights Hub (formerly MindSphere) cloud platform → Performance Insight dashboard app → digital twin of the full production line for what-if simulation
- Results (verified): Operators now get advance warning 12–36 hours before a potential failure; the digital twin work at Amberg cut defect rates by over 60% and reduced energy usage by 20% (WEF Global Lighthouse Network, 2023); simulation-driven optimization brought controller cycle time down to a target 8 seconds from an initial 11-second baseline
- Why It Matters: This is Slide 2's "Sensor → Edge → Enterprise" architecture running in production — not a hypothetical. Amberg is one of the WEF's officially recognized "Lighthouse Factories" for Industry 4.0, making it one of the most independently verified IIoT deployments in the world

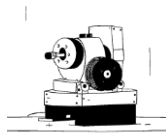


Trends, Strategic Risks & Next Steps



Edge AI + Agentic Systems

Local models enabling autonomous adjustments (beyond alerts). Expect hybrid training: cloud for models, edge for execution.



Digital Twins Mainstream

Twins enable what-if testing, faster diagnostics, and model-driven control loops across plant assets.



Security & Standards

Zero-trust architectures and standardized protocols become procurement requirements. Plan for identity, segmentation, and continuous monitoring.

Actionable next steps for engineering & ops leaders

1. Run an asset inventory & connectivity assessment (identify retrofit scope and data value).
2. Pilot one high-value use case (predictive maintenance or energy optimization) with measurable KPIs.
3. Design security-first architecture (segmentation, certificate-based device identity, OT/IT collaboration).
4. Plan scale: choose platforms that support edge deployment, open standards, and integration with MES/ERP.

Conclusion: IIoT is moving from passive monitoring to autonomous, self-correcting operations — engineers who pilot one high-value use case now, on a security-first architecture, will be positioned to scale as edge AI matures.

 References: Deloitte 2025 Smart Manufacturing Survey; Precedence Research IIoT Market Report 2026; IoT Analytics State of Enterprise IoT 2026.